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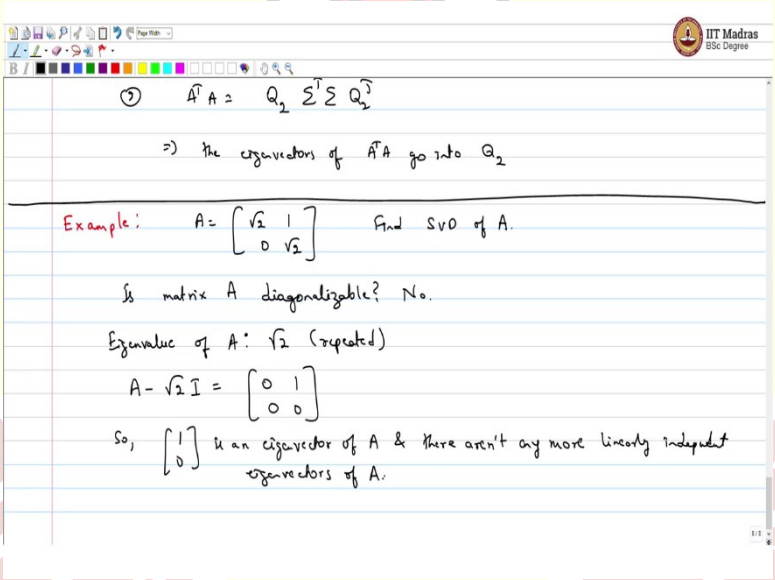
ONLINE DEGREE

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Example of SVD

Welcome to the final lecture of this week. In this lecture, we will not be introducing any new concepts, but we will just try to understand SVD through an example or a simple 2 cross 2 example, but all the concepts that we have learned will come into play. In particular, we will take an example and try to write it in the SVD form. But when you are, I mean, the example we are looking at is a square matrix, so the first thing to investigate is whether it is diagonalizable so we will have to check if it is diagonalizable.

How do we check that? By checking, if you get enough linearly independent eigenvectors. Once you do that, then we will go on to write it in the SVD form. And we will compute, we will show how to do this in detail.

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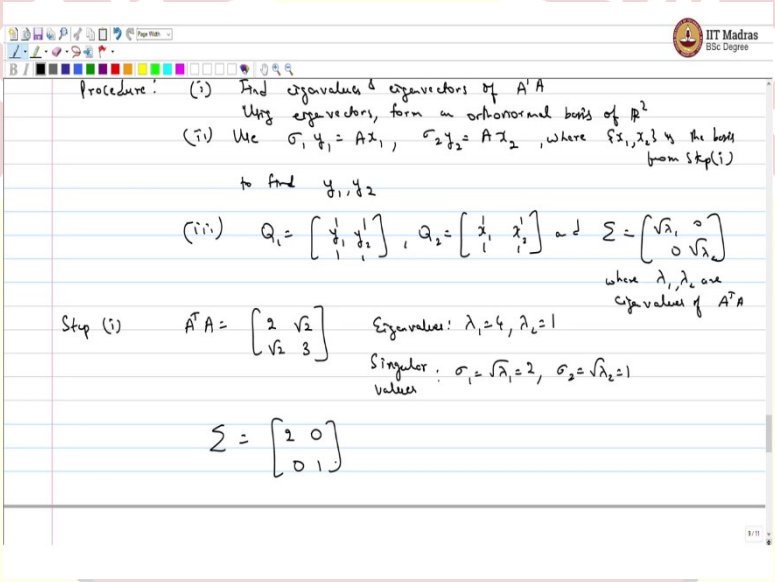
So, let us do an example to understand SVD. Even though it is a toy example with a 2 cross 2 matrix. And the concepts or the procedure of deriving the SVD of a given matrix should be clear from this example, let us take this matrix. The question is it is not symmetric, so I cannot apply spectral theorem to say it is diagonalizable. But there are, I mean, the task is to find SVD of A.

Now, for diagonalizable matrices, the SVD is the same, as the 1 given by spectral theorem so we may not work as hard. So, the first thing to check is, is matrix A they are diagonalizable. So let us check this. The answer is actually no.

So, eigenvalue is just √2 and this is repeated. But that is not enough to say that it is not diagonalizable. But because if you get two independent, linearly independent eigenvectors, you can diagonalize any matter, any 2 cross 2 matrix. So, let us check what is A - √2 I. This is A - λ I and this turns out to be 0 1 0 0.

And this, so, 1 0 is an eigenvector, and you do not get another linearly independent eigenvector. And there are not any more linearly independent eigenvector of A. So, it is clear, so you cannot diagonalize it.

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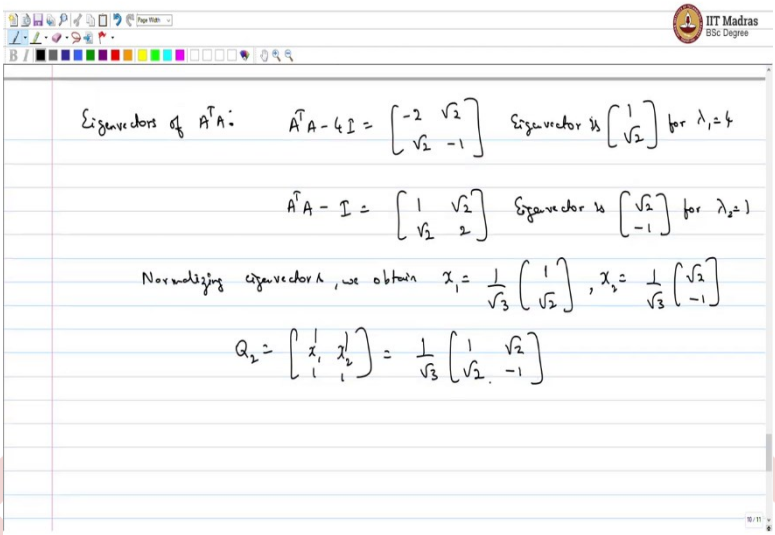


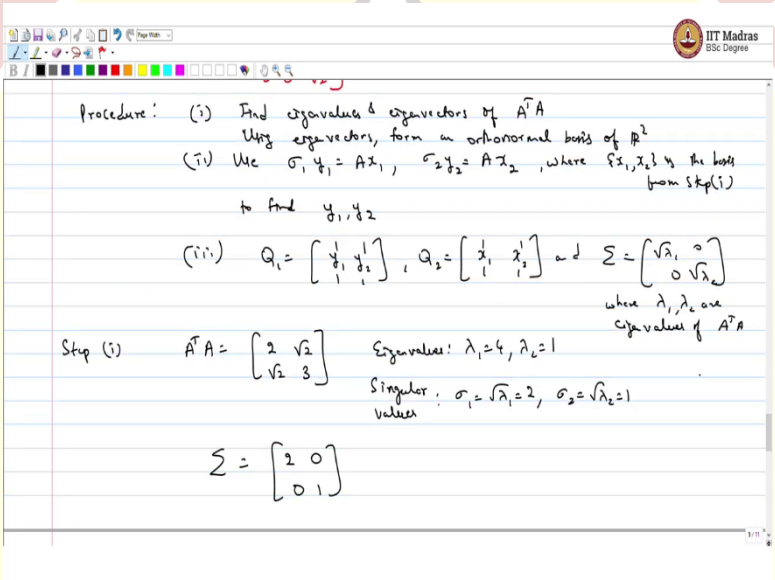
So, other reasonable thing to do is, to do the, let us do the SVD. So, finding the SVD of this matrix √2, this is 1, 0 √2. The procedure is find eigenvalues and eigenvectors of A transpose A using eigenvectors 1 and orthonormal basis. In this case it is R2.

Now, after this, use the fact that σ 1 when with y as defined in the proof Ax1, σ 2 y2 = Ax2 and x1 , x2 is the basis from step 1. Use this to find y1 , y2. And then you just have to put Q1 as y1, y2, Q2 as x1, x2, and σ as √λ 1, √λ 2. Where λ 1 , λ 2 are the eigenvalue of A transpose A.

So, this is the procedure. So let us go through this procedure fully. So, we are trying to find the SVD of this, so step 1. So, we know A transpose A is 2 √2, √2, 3. I mean, needless to say it is a real symmetric matrix. You can work this out on your own, it is easy to find the eigenvalues. They are λ 1 = 4, λ 2 = 1. So, the singular values, σ 1 is √λ 1 = 2, σ 2 = √λ 2 = 1. So, we are already ready to define or work out the matrix here.

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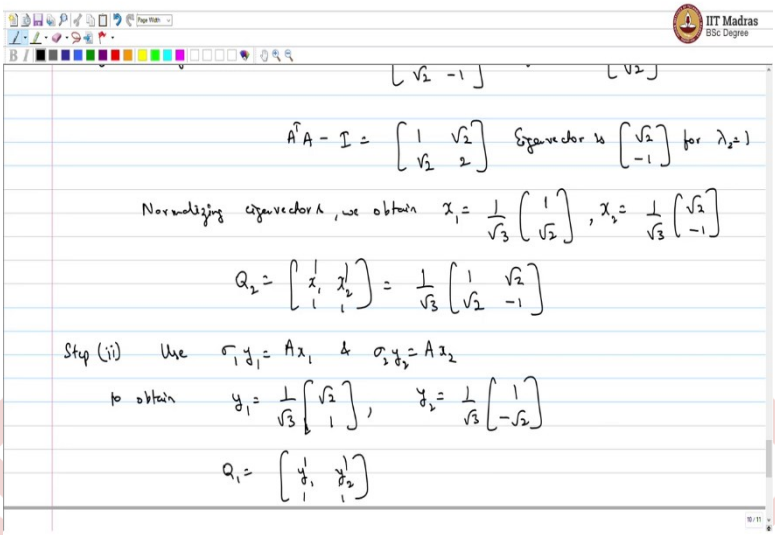


Now, the next step is finding the eigenvectors. Eigenvectors of A transpose A. So, the eigen values are 4 and 1, so the first thing is, A transpose A - 4 I, this turns out the I am going to write this for you, - 2 √2, √2 and - 1. And the other thing is A transpose A - I. So, A transpose A - I is 1 √2, √2, and 2.

Now, an eigenvector just by eyeballing eigenvector is, for this, it is 1 √2 for λ 1 = 4, the eigenvectors √2 and - 1 for λ 2 = 1. Now, we need to normalize, normalizing eigenvectors, we obtain there are two vectors, let me write it as x1 and x2. We will get x1 as just divide by the length 1 by √3, 1 , √2 and x2 = 1 by √3, square 2 and - 1.

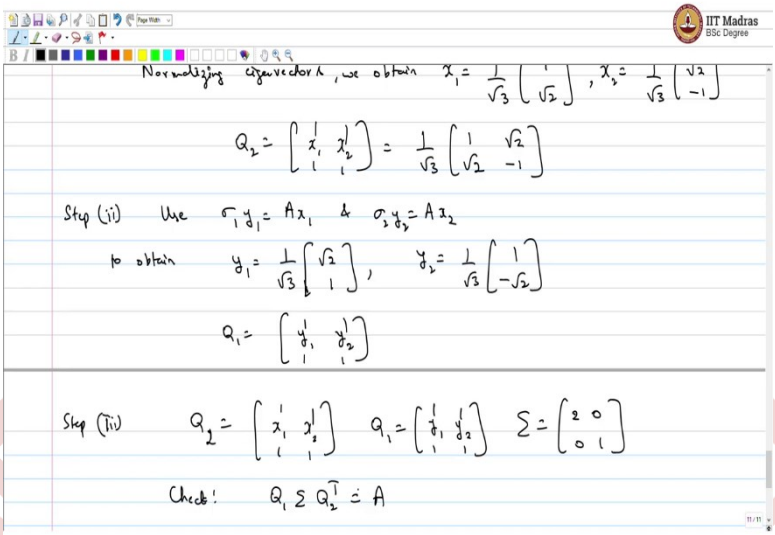
So, given these x1 and x2, what am I ready to define think about this. Given this, go back to step 2. Okay, Q2 can be defined now. Let me wait until, I will define Q2 here. So Q2 is just x1, x2. So, transferred to be just 1 by √3 is common, and we can discuss Q2, now, step 2.

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The Step 2 is we have to use σ 1, y1 = Ax1, and σ 2, y2 = Ax2. To obtain, you know everything about Ax1 and σ 1. So, you get y1 = 1 by √3, 1 by 2 =. So, I am ready to define Q1 now, y1, y2. By definition if you check the proof, writing σ 1, y1 = Ax1. By definition y1 and y2 will be unit length orthogonal.

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The last thing to check is, so what we have step 3. We have obtained Q1 is x1, x2, sorry Q2. Q1 is y1, y2 and σ is 2, 0, 0, 1. As a homework check Q1 σ Q2 transpose turns out to be check, out this is and read the case.

I mean, a very short summary before I wrap up, what did we learn in this lecture. We just applied the concepts that we had learned in the previous lecture, which is to understand how to decompose a matrix in the SVD form a given matrix, and we did this in the context of a simple 2 cross 2 matrix here.

And we also, at the outset, we had to do a sanity check. If a matrix is diagonalizable its eigen decomposition is its SVD form. However, the example that we took up in this lecture was not diagonalizable, which we verified and then we wrote it in the SVD form by computing the singular vectors and left and right eigenvectors of appropriate matrices.

Once we did that, we could write this given matrix in this SVD form. As a product of an orthogonal matrix × a diagonal matrix × another orthogonal matrix. And this orthogonal matrix the entries that go into this orthogonal matrix have some logical interpretations. They are the eigenvectors of either a given a matrix A, they are the eigenvectors of either A transpose A or A into A transpose, depending on whether they are on the left or the right of the diagonal matrix.